Checkpoint 6 Maths — Practice Test 1

Paper 1: 45 minutes · no calculator · Paper 2: 45 minutes · calculator allowed · Total 80 marks

Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 80

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 45 minutes — 40 marks

1 The school library counts its books. It has 2 thousands, 15 hundreds, 3 tens and 8 ones. **[1]**

How many books does the library have?

Working

Answer

2 A sequence counts back in steps of the same size. **[1]**

2.0, 1.4, 0.8, _____, _____

Write the next two numbers. Type them with a comma between them, for example 5, 3

3 **[1]**

(a) Work out 30 − 4 × 6 ÷ 3

Working

Answer

(b) Write one pair of brackets to make this calculation correct. **[1]**

5 + 3 × 4 − 2 = 30

4 A juice stall sells 1245 cups of juice in April. **[2]**
Each cup costs 24 pounds.

How much money does the stall take for these cups in April? Show your working.

Working

Answer ____ pounds

5 The juice stall has 912 oranges. The oranges are packed into bags of 16. **[2]**

How many bags can be filled? Show your working.

Working

Answer ____ bags

6 On a class trip to the zoo, the sea-lion show starts every 15 minutes. **[1]**
The parrot show starts every 20 minutes.
Both shows start at 10:00

What is the next time that both shows start together?

7 **[1]**

Which of these numbers is divisible by 9?

- ☐ A) 1236
- ☐ B) 4517
- ☐ C) 6024
- ☐ D) 2754

8 **[1]**

Which of these is a cube number?

- ☐ A) 16
- ☐ B) 27
- ☐ C) 49
- ☐ D) 50

9 In this sequence, each term is the position number squared, then subtract 1. **[1]**

Position	1	2	3	4
Term	0	3	8	15

What is the 9th term?

Working

Answer

10 Pupils make a pattern of books on the library display shelves. **[1]**

Position	1	2	3	4
Number of books	7	14	21	28

(a) How many books are there in position 20?

Working

Answer books

(b) Sara says, "One of the positions has exactly 150 books." **[1]**
Is Sara correct? Explain how you know.

11 On the zoo trip there are b boys and g girls. **[1]**

b + g = 42

b = 19. What is the value of g?

Working

Answer

12 At the zoo, the temperature inside the penguin house is −3 °C. **[1]**

The temperature inside the reptile house is 28 °C.

What is the difference between the two temperatures?

Working

Answer °C

13 **[1]**

What is the value of the digit 6 in the number 4.316?

Working

Answer

14 **[1]**

Work out 45 ÷ 1000

- ☐ A) 0.45
- ☐ B) 4.5
- ☐ C) 0.045
- ☐ D) 0.0045

15 **[1]**

(a) Ola says, "20/32 written in its simplest form is 10/16."
Is Ola correct? Tick (✓) Yes or No and explain your answer.

(b) Work out 3/5 of 45 **[1]**

Working

Answer

16 In the school library, 2/5 of the books are fiction and 1/3 of the books are non-fiction. **[2]**
The rest of the books are magazines.

What fraction of the books are magazines? Show your working and give your answer in its simplest form.

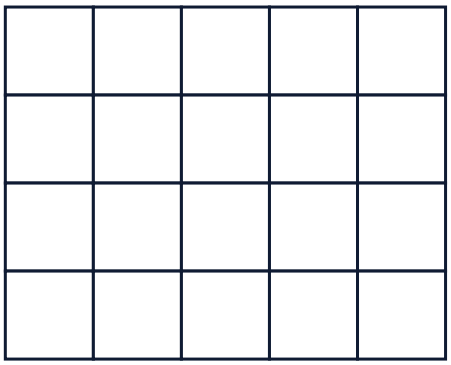
17 **[1]**

Work out 2/9 × 4

18 **[1]**

Write <, > or = in each box.
0.7 ☐ 3/4
40% ☐ 2/5

19 **[1]**



Shade 35% of this grid.

20 **[1]**

Work out 7.3 − 2.86

Working

Answer

21 The juice stall makes mango-orange juice. **[1]**

It mixes mango juice and orange juice in the ratio 2 : 5

The stall uses 6 cups of mango juice. How many cups of orange juice does it use?

Working

Answer cups

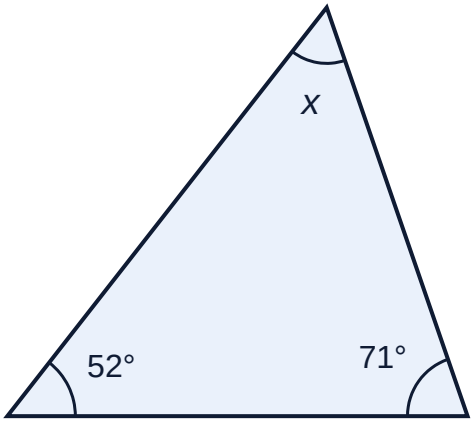
22 The class trip to the zoo lasts 3.75 hours. **[1]**

3.75 hours = 3 hours and how many minutes?

Working

Answer ____ minutes

23 **[1]**



Work out the size of angle x.

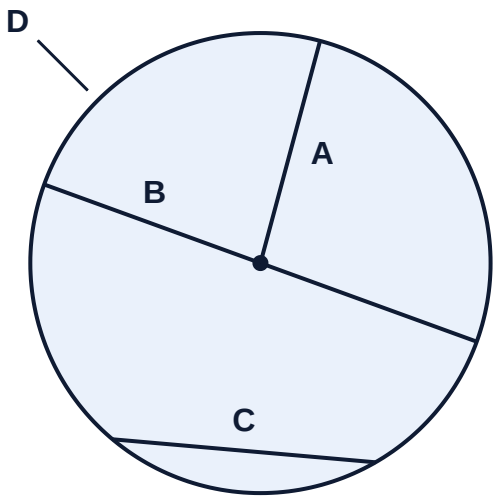
Working

Answer ____ °

24 **[1]**

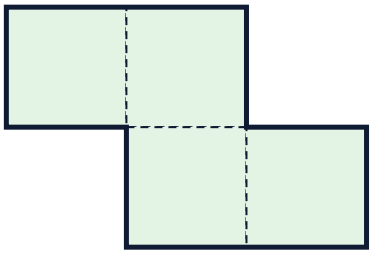
Sketch a quadrilateral that has exactly one pair of parallel sides and two right angles.
Write the name of your shape.

25 Look at the circle. The dot is the centre. **[1]**



Which letter shows a radius of the circle?

26 A garden path is made from paving slabs in this shape. **[1]**

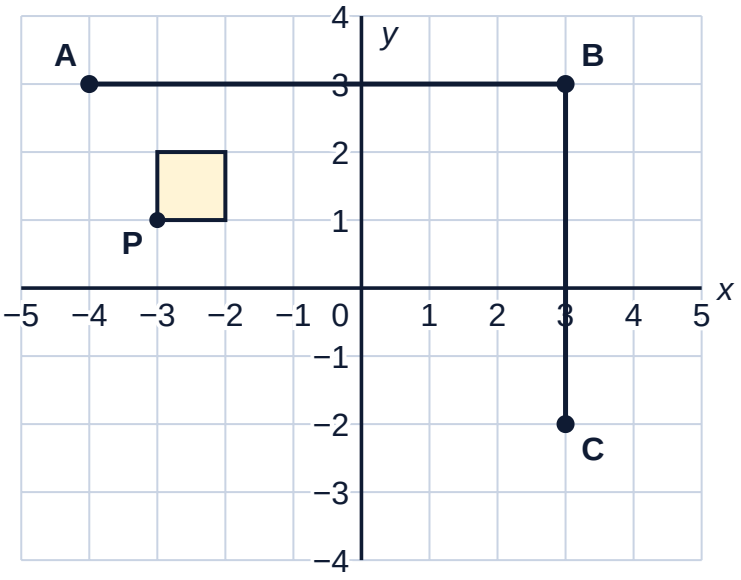


What is the order of rotational symmetry of the shape?

Working

Answer

27 A garden plot plan is drawn on a coordinate grid. A, B and C are three corners of a rectangular plot. The square is a shed. P is one corner of the shed. **[1]**



(a) Write the coordinates of point A.

(b) D is the fourth corner of the rectangle ABCD. Write the coordinates of D. **[1]**

(c) The shed is moved 5 squares to the right and 2 squares down. Write the new coordinates of corner P. **[1]**

28 A bag has 6 red counters and 2 blue counters. Ali takes one counter without looking. **[1]**

Which word best describes the chance that Ali takes a blue counter?

- ☐ A) unlikely
- ☐ B) impossible
- ☐ C) even chance
- ☐ D) likely

29 A spinner has eight equal sections numbered 1 to 8. Rana spins it once. **[1]**

Explain why the events "the number is odd" and "the number is a multiple of 4" are mutually exclusive.

30 Seven pupils write down how many library books they borrowed this term. **[1]**

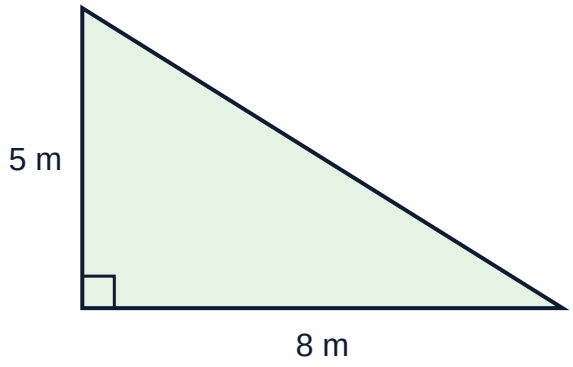
3, 5, 2, 5, 8, 4, 1

What is the median number of books?

Working

Answer

31 A flower bed in the garden is a right-angled triangle. **[1]**



Work out the area of the flower bed.

Working

Answer m^2

32 Omar says, "Every number that is divisible by 3 is also divisible by 6." **[1]**

Give an example to show that Omar is wrong. Explain your example.

Paper 2 — 45 minutes — 40 marks

1 At sports day, seven pupils from Class 6A run 100 m. The table shows their times. **[1]**

Name	Adam	Karim	Youssef	Omar	Ziad	Hamza	Seif
Time (seconds)	15.42	14.87	16.05	15.20	14.87	15.80	15.66

(a) Round Seif's time to the nearest tenth of a second.

Working

Answer ____ seconds

(b) What is the mode of the times? **[1]**

Working

Answer ____ seconds

(c) What is the median time? **[1]**

Working

Answer ____ seconds

2 At sports day, seven pupils from Class 6A run 100 m. The table shows their times. **[2]**

Name	Adam	Karim	Youssef	Omar	Ziad	Hamza	Seif
Time (seconds)	15.42	14.87	16.05	15.20	14.87	15.80	15.66

Work out the mean time. Show your working.

Working

Answer ____ seconds

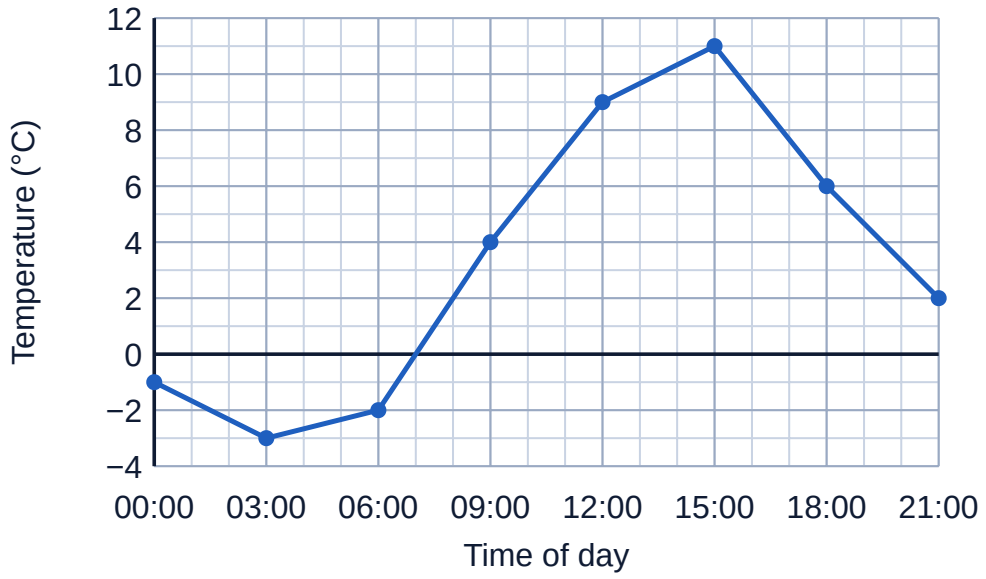
3 At sports day, seven pupils from Class 6A run 100 m. The table shows their times. **[2]**

Name	Adam	Karim	Youssef	Omar	Ziad	Hamza	Seif
Time (seconds)	15.42	14.87	16.05	15.20	14.87	15.80	15.66

For Class 6B the mean time is 15.90 seconds and the range is 0.62 seconds.

Compare the times of Class 6A and Class 6B. Write two different comparisons.

4 The weather chart shows the temperature in a town in Sinai, measured every 3 hours on one day in January. **[1]**



(a) What was the temperature at 09:00?

Working

Answer ____ °C

(b) What is the difference between the highest and the lowest temperature on the chart? **[1]**

Working

Answer ____ °C

(c) Use the graph to estimate the temperature at 08:00. **[1]**

Working

Answer ____ °C

(d) Explain why a line graph is a good way to show this data. **[1]**

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5 The juice stall shows this price list. **[2]**

Juice	Price of one cup
Orange	18.50 pounds
Mango	22.75 pounds
Guava	15.25 pounds

Huda buys 3 cups of mango juice and 2 cups of guava juice. She pays with a 200-pound note.
How much change does she get? Show your working.

Working

Answer ____ pounds

6 **[1]**

A jug holds 2.4 litres of orange juice. It is poured equally into 8 cups.
How much juice is in each cup?

Working

Answer ____ litres

7 **[1]**

The stall uses 6 mangoes to make 4 cups of mango juice.
How many mangoes does it need to make 18 cups?

Working

Answer ____ mangoes

8 **[1]**

On Friday the stall sells 240 cups of juice. 35% of them are orange juice.
How many cups of orange juice does it sell?

Working

Answer ____ cups

9 152 pupils and 9 teachers go on a class trip to the zoo. **[1]**

(a) Each bus can carry 48 people. What is the smallest number of buses they need?

Working

Answer ____ buses

10 152 pupils and 9 teachers go on a class trip to the zoo. **[2]**
Zoo tickets cost 85 pounds for each pupil and 120 pounds for each teacher.

(b) Work out the total cost of the tickets. Show your working.

Working

Answer ☐ pounds

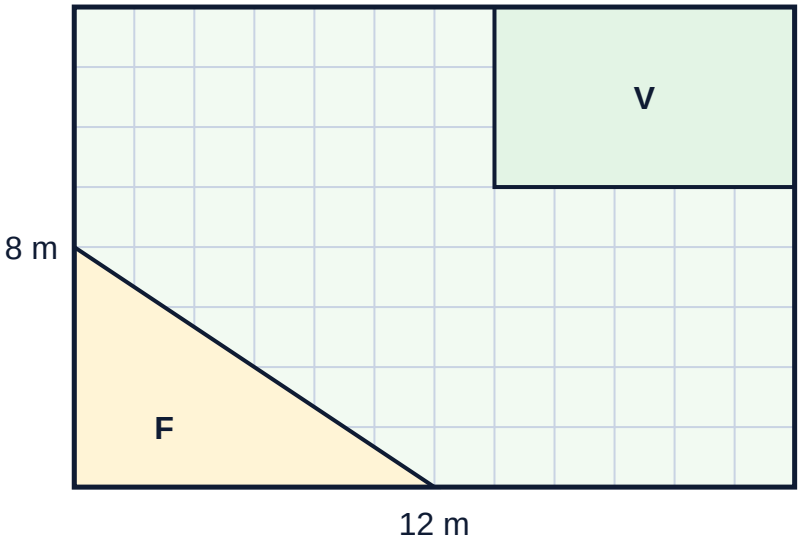
11 **[1]**

The bus journey home takes 1.4 hours.
1.4 hours = 1 hour and how many minutes?

Working

Answer ☐ minutes

12 The plan shows a rectangular school garden plot on a square grid. Each square is 1 m by 1 m. **[1]**



F = flower bed, V = vegetable bed, the rest is grass

(a) Work out the area of the flower bed F.

Working

Answer ____ m^2

(b) Work out the area of the grass. Show your working. **[2]**

Working

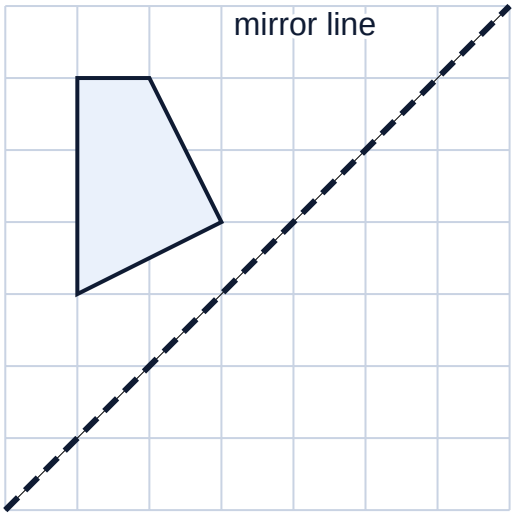
Answer ____ m^2

(c) The vegetable bed is shared between tomatoes and beans in the ratio 3 : 2 **[1]**
What area is used for tomatoes?

Working

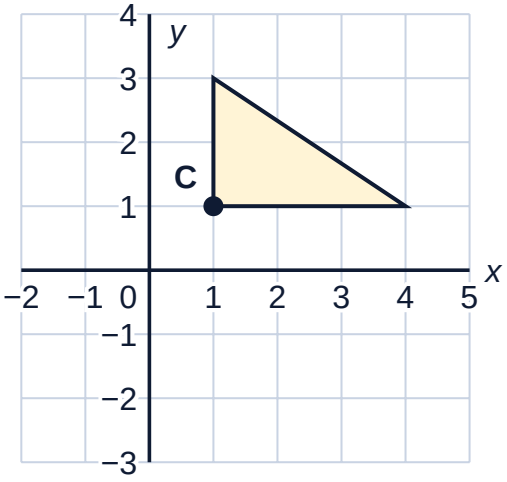
Answer ____ m^2

13 **[1]**



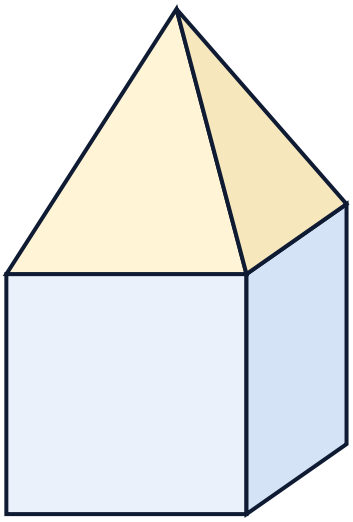
Draw the reflection of the shape in the mirror line.

14 **[1]**



Rotate the triangle 90° clockwise about the vertex C.
Draw the new triangle on the grid.

15 This shape is made from a cube with a square-based pyramid on top. **[1]**



How many faces does the shape have?

Working

Answer ____ faces

16 **[1]**

Sketch a net of a square-based pyramid.

17 **[1]**

Which sentence describes capacity?

- ☐ A) The amount of space that a solid object takes up
- ☐ B) The amount that a container can hold when it is full
- ☐ C) The total area of all the faces of a 3D shape
- ☐ D) The distance all the way around a 2D shape

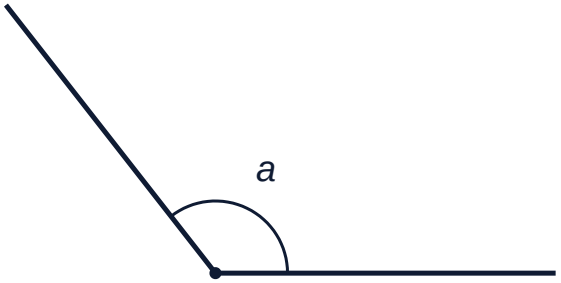
18 Books are delivered to the library in a box shaped like a cuboid. It is 30 cm long, 20 cm wide and 10 cm high. **[2]**

Work out the surface area of the box. Show your working.

Working

Answer ____ cm^2

19 **[1]**



Estimate the size of angle a.

- ☐ A) 128°
- ☐ B) 38°
- ☐ C) 52°
- ☐ D) 152°

20 **[1]**

Hassan wants to draw a circle with a diameter of 9 cm using compasses.
How far apart should he set the point and the pencil of the compasses? Explain your answer.

21 At the zoo shop there is a game with a spinner showing four animals. Mona spins it 100 times. **[1]**
These are her results.

Animal	lion	zebra	monkey	giraffe
Frequency	12	25	9	54

(a) Mona's friend says, "Each animal is equally likely."
Do Mona's results support this? Explain your answer.

(b) Why is it better to spin the spinner 100 times than 10 times? **[1]**

- ☐ A) It makes the spinner fair
- ☐ B) Each animal will then come up the same number of times
- ☐ C) The results give a more reliable estimate of the probability
- ☐ D) The spinner will land on giraffe every time

22 At the zoo shop there is a game with a spinner showing four animals. Mona spins it 100 times. **[1]**
These are her results.

Animal	lion	zebra	monkey	giraffe
Frequency	12	25	9	54

Mona spins the spinner another 200 times. About how many times should she expect it to land on zebra?

Working

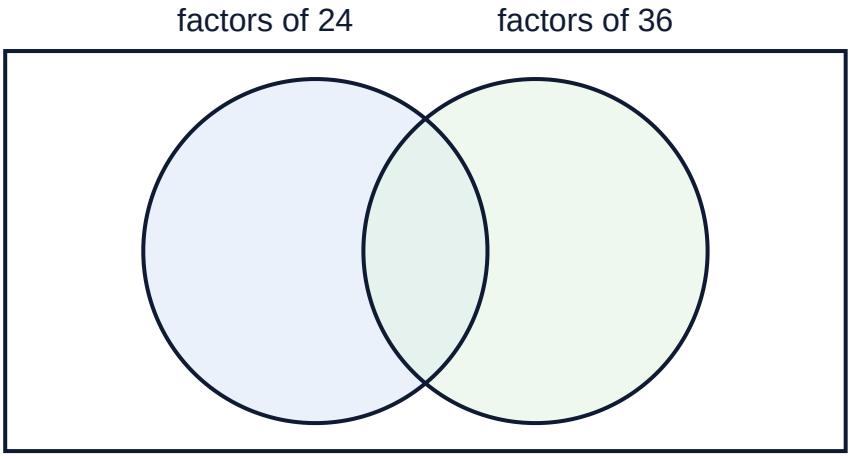
Answer ____ times

23 Pupils collect different kinds of data on sports day. **[1]**

Which of these is continuous data?

- ☐ A) the number of medals each class wins
- ☐ B) each pupil's favourite sport
- ☐ C) the number of pupils in each race
- ☐ D) the time each pupil takes to run 100 m

24 **[1]**



Write the numbers 5, 8, 9, 12 and 18 in the correct places on the Venn diagram.

25 7 litres of juice are shared equally between 4 jugs. **[1]**

How many litres are in each jug? Give your answer as a mixed number or a decimal.

Working

Answer ____ litres

26 A sequence starts at 2.5 and goes up by 0.75 each time. **[1]**

2.5, 3.25, 4, 4.75, …

Is 10 a term in this sequence? Explain how you know.